

Kapil Kumar

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Professional Summary

Python engineer specialising in agentic AI architectures – designing and orchestrating multi-agent LLM workflows (LangChain, LangGraph) with tool-use, memory, and planning, integrated with REST APIs and RAG-based knowledge retrieval. Builds, tests, and deploys production agent pipelines with dedicated evaluation/debugging tooling, guardrails, and monitoring for reliable, enterprise-grade AI systems on Docker and Azure.

Technical Skills

Programming: Python (OOP, data structures, clean/documented code), Java, SQL, TypeScript

Agentic AI: LangChain, LangGraph (multi-agent orchestration, task planning, memory, tool-use, checkpointed state), Prompt Engineering, Agent Evaluation & Debugging Pipelines

LLM & Retrieval: RAG Architecture, pgvector (vector retrieval), Embeddings (sentence-transformers), Groq API, Knowledge-base Integration

Tool & API Integration: REST APIs, Webhooks, External Tool Augmentation, ElevenLabs Voice AI, Twilio

API Frameworks: FastAPI, Flask, Django

Databases: PostgreSQL, DuckDB, MongoDB, Redis, Parquet

Safety & Reliability: Query-level guardrails (sqlglot AST validation), fallback mechanisms, JWT Auth

Cloud, Deployment & CI/CD: Azure (Blob Storage, VM, PostgreSQL), Docker, Docker Compose, GitHub Actions CI/CD

Tools: Git, GitHub, Make.com, n8n, Postman

Experience

Associate Software Engineer

Dec 2025 – Present

Aviara Labs

Noida

- Developed low-latency, full-duplex AI voice agents (ElevenLabs) handling scheduling actions and live call-transfers, sustaining 95% uptime across 50,000+ monthly calls and 100k+ minutes.
- Designed and orchestrated multi-agent LLM workflows (LangGraph) for a healthcare scheduling platform – multi-turn conversation state, tool-use, and context-aware planning across FastAPI and Django services – cutting appointment scheduling time ~60% (20 to 8 minutes).
- Built **VoiceQA** – an agent evaluation and debugging pipeline that fires scenario-driven test runs, ingests transcripts via webhook, and scores agent performance through a two-stage LLM pipeline (free-form analysis → structured pass/fail scoring), persisting results in PostgreSQL and surfacing pass-rate/trend reports via REST and CLI.
- Integrated external tools and REST APIs (Make.com, n8n) to augment agent capabilities across backend systems; deployed services via Docker with GitHub Actions CI/CD and JWT-secured API access.

Apprentice

Oct 2025 – Dec 2025

Air India Limited

Gurugram

- Collected, cleaned, and analysed complex business and financial datasets from multiple sources using Python (Pandas), SQL, and Excel to support data-driven decision-making.

Projects

ApplyPilot | *FastAPI, PostgreSQL + pgvector, Redis, Celery, Next.js, Typst, LaTeX*

2026

- Built a human-in-the-loop job-application copilot: job discovery across 9 keyless ATS connectors (incl. a country-scoped Workday connector), an Overleaf-style résumé editor (Typst + LaTeX), and JD-driven tailoring with explainable 8-dimension match scoring.
- Designed a deterministic provenance verifier that checks every rewritten bullet (rule- or LLM-generated) against source evidence and blocks unsupported numbers, tools, or employers from export; a compatibility analyzer re-parses the compiled PDF with two independent parsers.
- Backed by 623 tests, including a red-team suite for the verifier, with GitHub Actions CI against real Postgres and Redis; deployed on Azure and Vercel.
- Live: applypilot.kapilp.tech

ExcelMind | *FastAPI, LangGraph, pgvector, Groq LLM, DuckDB, React*

2025

- Built a LangGraph-based agentic copilot with persistent chat memory (thread-scoped state), a /slash-command tool system, and upload-triggered workflows for multi-step reasoning over user data.
- Implemented RAG-based retrieval with pgvector embeddings and a sqlglot AST-level query guard as a safety layer between the LLM agent and the database, preventing unsafe or unauthorized query execution.
- Re-engineered the data pipeline (Polars + DuckDB), achieving a ~50× faster parse and sub-100 ms queries over multi-million-row datasets; deployed via Docker Compose on Azure VM.
- Live: excelmind.kapilp.tech

Contract Intelligence API | *FastAPI, RAG, LLMs, Docker, Redis, SSE* **Aug 2025**

- Built a production GenAI service for document analysis: PDF ingestion, RAG-based question answering across multiple LLM providers, real-time SSE streaming, and an LLM-based risk-audit engine with regex fallback for reliable output during rate limiting.
- GitHub: github.com/Kapilkumar16/contract-intelligence-api

Education

College of Engineering Roorkee

Sep 2021 – June 2025

Bachelor of Technology in Computer Science and Engineering

Roorkee, India

Certifications

Django Full Stack Development – Udemy

Data Visualization – TATA (Forage)